

DEPARTMENT OF HOMELAND SECURITY
Customs and Border Protection
Office of Acquisition

TECHNICAL SPECIFICATION
TS-2024-HALE-SENSOR-001

Multi-Band Electro-Optical / Infrared Sensor Payload Suite
for High-Altitude Long-Endurance Unmanned Aerial Systems

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This document establishes the technical requirements, interface definitions, performance parameters, environmental qualification criteria, and logistics support data for the Multi-Band EO/IR Sensor Payload Suite intended for integration on High-Altitude Long-Endurance (HALE) unmanned aerial systems supporting persistent border surveillance missions.

WARNING: This document contains technical data whose export is restricted by the Arms Export Control Act (Title 22, U.S.C., Sec 2751 et seq.) or the Export Administration Act of 1979, as amended (Title 50, U.S.C., App. 2401 et seq.). Violations of these export laws are subject to severe criminal penalties.

1. SCOPE

1.1 Purpose. This Technical Specification (TS) defines the functional, performance, interface, environmental, and logistics requirements for a Multi-Band Electro-Optical / Infrared (EO/IR) Sensor Payload Suite (hereafter “the Payload”) suitable for installation on Government-furnished High-Altitude Long-Endurance Unmanned Aerial Systems (HALE-UAS). The Payload shall provide day/night, all-weather detection, recognition, and identification capability against personnel, vehicles, and low-observable maritime surface targets at stand-off ranges consistent with HALE operating altitudes of 45,000 to 65,000 feet Mean Sea Level (MSL).

1.2 Applicability. This TS applies to all contractor deliverables under Solicitation HSCBP-24-C-00087 and any subsequent production, sustainment, or modification contracts that reference this document. Compliance with every shall-statement is mandatory unless a formal deviation or waiver has been approved in writing by the Contracting Officer’s Technical Representative (COTR).

1.3 System Overview. The Payload consists of a stabilized multi-sensor turret, onboard processing unit, data-link interface electronics, power conditioning module, and associated cabling and mounting hardware. The turret houses a high-definition visible-band camera, a mid-wave infrared (MWIR) imager, a short-wave infrared (SWIR) imager, a laser rangefinder / designator, and an inertial measurement unit (IMU) for geo-location. The system is required to operate continuously for mission durations up to 30 hours without ground intervention other than periodic command uplinks.

2. APPLICABLE DOCUMENTS

2.1 Government Documents. The following documents form a part of this specification to the extent specified herein. In the event of conflict between the text of this specification and the references cited herein, the text of this specification takes precedence.

Document Number	Title	Revision / Date
MIL-STD-810H	Environmental Engineering Considerations and Laboratory Tests	31 Jan 2019
MIL-STD-461G	Requirements for the Control of Electromagnetic Interference	11 Dec 2015
MIL-STD-704F	Aircraft Electric Power Characteristics	12 Mar 2004
MIL-HDBK-217F	Reliability Prediction of Electronic Equipment	Notice 2, 28 Feb 1995
STANAG 4586	Standard Interfaces of UAV Control System for NATO UAV Interoperability	Ed. 4
ICD-GPS-200	Navstar GPS Space Segment / Navigation User Interfaces	Rev N
CBP-ARCH-2022-03	CBP Unmanned Systems Architecture Framework	Rev B, 2022

2.2 Non-Government Documents. The following industry standards are invoked as requirements where cited.

Document Number	Title	Revision
IEEE 802.3	Ethernet Standard (1000BASE-T / 10GBASE-T)	2022
ARINC 818-3	Avionics Digital Video Bus	2019
RTCA DO-160G	Environmental Conditions and Test Procedures for Airborne Equipment	Dec 2010
SAE AS9100D	Quality Management Systems – Requirements for Aviation	2016

3. REQUIREMENTS

3.1 System Performance Requirements.

The Payload shall meet or exceed the performance values listed in Table 3-1 under the environmental conditions defined in Section 3.5. All optical performance values are specified at the sensor aperture and include the effects of the protective window and any internal relay optics.

Table 3-1. Key Performance Parameters

Parameter	Requirement	Unit	Verification Method
Visible-band horizontal FOV (narrow)	0.35 ± 0.02	degrees	Analysis + Test
Visible-band horizontal FOV (wide)	28.0 ± 1.0	degrees	Analysis + Test
MWIR horizontal FOV (narrow)	0.45 ± 0.03	degrees	Analysis + Test
MWIR NEdT (at f/4, 300 K scene)	≤ 25	mK	Test
SWIR detection range (man-sized target, 50 % P _d)	≥ 18	km	Analysis + Demo
MWIR recognition range (vehicle, NATO criteria)	≥ 22	km	Analysis + Demo
Laser rangefinder maximum range	≥ 40	km	Test
Laser rangefinder accuracy (1σ)	≤ 2.0	m	Test
Stabilization residual (1σ , high-frequency)	≤ 5	μ rad	Test
Geo-location accuracy (CEP50 at 50 km slant range)	≤ 15	m	Analysis + Test
Continuous mission endurance (powered)	≥ 30	hours	Analysis
Time to first image after power-on	≤ 90	seconds	Test
Maximum power draw (steady-state, all sensors)	≤ 850	W	Test
Peak power draw (laser fire + all sensors)	$\leq 1\,400$	W	Test
Mass (turret + electronics, dry)	≤ 68	kg	Inspection

3.1.1 Detection / Recognition / Identification Criteria. Target discrimination ranges shall be calculated in accordance with the NATO STANAG 4347 methodology using a 50 % probability of detection (P_d) and a false-alarm rate not exceeding 10^{-4} per square kilometer searched per hour. Atmospheric transmission shall be modeled using MODTRAN 5 with a mid-latitude summer, 23 km visibility profile unless otherwise directed by the Government.

3.2 Interface Requirements.

3.2.1 Electrical Power. The Payload shall accept aircraft power conforming to MIL-STD-704F, Category A, 28 VDC nominal. Inrush current shall not exceed 80 A for more than 50 ms. The power conditioning module shall provide isolated outputs of +28 V, +15 V, -15 V, and +5 V with ripple not exceeding 50 mV p-p.

3.2.2 Data Interfaces. Primary command and control shall be via dual redundant 1000BASE-T Ethernet ports conforming to IEEE 802.3. Video streams shall be provided as ARINC 818-3 (Fibre Channel) at 3.1875 Gbps and as compressed H.265 over Ethernet. A discrete RS-422 serial channel shall be provided for legacy autopilot health monitoring at 115 200 baud, 8N1.

3.2.3 Mechanical Interface. The turret shall mount to a Government-furnished hard-point using a 12-bolt pattern defined in ICD-HALE-TURRET-07 Rev B. Maximum protrusion below the aircraft skin shall not exceed 420 mm. Vibration isolators shall attenuate aircraft structure-borne vibration by a minimum of 20 dB above 50 Hz.

3.3 Reliability, Maintainability, and Availability.

The Payload Mean Time Between Critical Failures (MTBCF) shall be no less than 2 500 operating hours when calculated in accordance with MIL-HDBK-217F, Part Stress Analysis, Ground Mobile environment, temperature 40 °C. Mean Time To Repair (MTTR) at organizational level shall not exceed 45 minutes for any Line-Replaceable Unit (LRU). Built-in-test (BIT) coverage shall detect at least 95 % of all mission-critical failures.

4. LOGISTICS AND PROVISIONING DATA

4.1 Bill of Materials – Critical Items. Table 4-1 lists the critical LRUs, subassemblies, and selected piece-parts required for initial provisioning and two-year sustainment. Part numbers shown are the contractor's internal designators; corresponding National Stock Numbers (NSNs) will be assigned after first-article approval. Unit prices are Rough-Order-of-Magnitude (ROM) estimates in Fiscal Year 2024 constant dollars and are provided for planning purposes only.

Contractors shall maintain configuration control of every item listed. Any change to form, fit, function, or part number requires Government approval via an Engineering Change Proposal (ECP) prior to incorporation.

Table 4-1. Critical Items Bill of Materials (BOM) and Provisioning Data

Item No.	Part Number	Nomenclature	Qty per System	Unit Price (USD)	MTBF (hrs)	Cage Code
001	QEO-TUR-1000-A	Stabilized Multi-Sensor Turret Assembly	1	412 500	4 200	1X9K2
002	QEO-VIS-2048-C	Visible-Band CMOS Camera Module, 2048×1536	1	28 750	12 000	1X9K2
003	QEO-MWIR-640-B	MWIR InSb FPA Module, 640×512, f/4	1	95 200	8 500	1X9K2
004	QEO-SWIR-1280-A	SWIR InGaAs Camera, 1280×1024	1	67 800	9 100	3B7F4
005	QEO-LRF-40K-D	Eye-Safe Laser Rangefinder / Designator, 40 km	1	54 300	6 800	1X9K2
006	QEO-IMU-300-A	Tactical-Grade IMU / AHRS	1	18 900	15 000	5T2M8
007	QEO-PCU-850-B	Power Conditioning Unit, 850 W continuous	1	12 400	18 000	1X9K2
008	QEO-VPU-10G-C	Video Processing Unit, dual 10 GbE + ARINC 818	1	31 600	11 200	1X9K2
009	QEO-WIN-IR-VIS	Multi-Spectral Protective Window Assembly	1	8 750	N/A	1X9K2
010	QEO-GIM-3AX-A	Three-Axis Direct-Drive Gimbal Mechanism	1	47 200	7 500	1X9K2
011	QEO-CAB-HARN-01	Primary Power & Data Harness Set	1	4 850	N/A	1X9K2
012	QEO-CAB-HARN-02	Video & Discrete Interface Harness	1	3 920	N/A	1X9K2
013	QEO-MNT-ADP-12	12-Bolt Aircraft Mount Adapter Plate	1	2 150	N/A	1X9K2
014	QEO-ISO-VIB-20	Vibration Isolator Set (4 ea)	1	1 680	N/A	7K3P1
015	QEO-FAN-48V-A	Cooling Fan Assembly, 48 VDC	2	420	25 000	1X9K2
016	QEO-FILT-EMI-28	EMI Filter Module, 28 VDC input	1	890	40 000	1X9K2
017	QEO-CONN-D38999	Circular Connector, D38999 Series III, 22-55	8	185	N/A	1X9K2
018	QEO-CONN-RJ45-M	Ruggedized RJ45 Ethernet Connector	4	95	N/A	1X9K2
019	QEO-OPT-LENS-VIS	Visible-Band Zoom Lens Assembly, 15–450 mm	1	14 600	10 000	4R8T2
020	QEO-OPT-LENS-MW	MWIR Continuous-Zoom Lens, 30–600 mm	1	38 900	8 000	4R8T2
021	QEO-OPT-LENS-SW	SWIR Fixed Focal Length Lens, 250 mm	1	11 200	12 000	4R8T2
022	QEO-FPA-COOL-A	Closed-Cycle Cryocooler for MWIR FPA	1	22 500	5 500	1X9K2
023	QEO-PSU-5V-15	Secondary PSU, +5 V / ±15 V outputs	1	1 950	22 000	1X9K2
024	QEO-BIT-MOD-A	Built-In-Test / Health Monitoring Module	1	3 400	20 000	1X9K2
025	QEO-GPS-OEM-7	Multi-Constellation GNSS Receiver Module	1	2 780	30 000	6N4W9
026	QEO-SSD-2TB-A	Rugged Solid-State Recorder, 2 TB	1	4 650	50 000	1X9K2
027	QEO-HEAT-SINK-M	Primary Heat Sink / Cold-Plate Assembly	1	1 120	N/A	1X9K2
028	QEO-SEAL-ORING-A	Environmental Seal Kit (O-rings & gaskets)	1	310	N/A	1X9K2
029	QEO-SW-FW-3.2	Sensor Firmware Load, Version 3.2.1	1	0 (NRE)	N/A	1X9K2
030	QEO-SW-GS-2.1	Ground-Station Control Software License	1	18 000	N/A	1X9K2
031	QEO-CAL-TGT-IR	IR Calibration Blackbody Target (lab)	1	6 400	N/A	2P5L7
032	QEO-CAL-TGT-VIS	Visible Collimator / Resolution Target	1	3 850	N/A	2P5L7
033	QEO-TOOL-KIT-A	Organizational Maintenance Tool Kit	1	2 900	N/A	1X9K2
034	QEO-SPARES-KIT1	30-Day Spares Kit (consumables & high-fail)	1	9 750	N/A	1X9K2
035	QEO-TRANS-CASE	Reusable Shipping / Transit Case	1	1 450	N/A	1X9K2
036	QEO-DOC-SET-A	Technical Manuals & IETM Package	1	4 200	N/A	1X9K2
037	QEO-TRN-PKG-B	Operator & Maintainer Training Package	1	12 500	N/A	1X9K2
038	QEO-FPGA-XC7-A	Main Processing FPGA, Xilinx Kintex-7	2	1 280	N/A	1X9K2
039	QEO-DDR4-8G	DDR4 SODIMM, 8 GB, industrial temp	4	185	N/A	1X9K2
040	QEO-OSC-100M	OCXO 100 MHz Reference Oscillator	1	420	80 000	8M2Q3

Item No.	Part Number	Nomenclature	Qty per System	Unit Price (USD)	MTBF (hrs)	Cage Code
041	QEO-ADC-14B	14-bit ADC Module, 125 MSPS	3	310	N/A	1X9K2
042	QEO-DAC-16B	16-bit DAC Module for laser control	1	275	N/A	1X9K2
043	QEO-LNA-MWIR	Low-Noise Amplifier, MWIR readout	1	890	N/A	1X9K2
044	QEO-TEC-CTRL	Thermo-Electric Cooler Controller Board	1	640	15 000	1X9K2
045	QEO-BLDC-DRV	Brushless DC Motor Driver (gimbal axes)	3	520	18 000	1X9K2
046	QEO-ENC-26B	26-bit Absolute Optical Encoder	3	1 150	25 000	9K4R5
047	QEO-BRG-CER	Ceramic Hybrid Bearing Set (gimbal)	6	380	N/A	1X9K2
048	QEO-SLIP-RING	Fiber-Optic Rotary Joint / Slip Ring	1	7 800	6 000	1X9K2
049	QEO-PRESS-SNS	Internal Pressure / Humidity Sensor	1	145	40 000	1X9K2
050	QEO-TEMP-SNS	Distributed Temperature Sensor Array	1	220	50 000	1X9K2
051	QEO-FUSE-30A	Power Input Fuse, 30 A slow-blow	2	18	N/A	1X9K2
052	QEO-TVS-28V	Transient Voltage Suppressor, 28 V bus	4	12	N/A	1X9K2
053	QEO-CAP-BUS	Bus Capacitor Bank, 4700 μ F / 50 V	1	95	N/A	1X9K2
054	QEO-IND-EMI	Common-Mode Choke, EMI suppression	2	35	N/A	1X9K2
055	QEO-PCB-MAIN	Main Processor PCB Assembly (loaded)	1	4 850	N/A	1X9K2
056	QEO-PCB-VID	Video Capture & Compression PCB	1	3 200	N/A	1X9K2
057	QEO-PCB-PWR	Power Distribution PCB Assembly	1	1 680	N/A	1X9K2
058	QEO-PCB-IO	Discrete I/O & Serial Interface PCB	1	980	N/A	1X9K2
059	QEO-FW-LOADER	Firmware Loader / JTAG Adapter	1	350	N/A	1X9K2
060	QEO-TEST-JIG	Factory Acceptance Test Jig (special tooling)	1	22 000	N/A	1X9K2

Note: Items marked "N/A" under MTBF are non-electronic or consumable items for which a statistical failure rate is not applicable. Cage Codes are fictional placeholders for document-generation purposes.

4.2 Recommended Initial Spares Allocation.

Based on the predicted failure rates and the planned annual flying-hour program of 2 400 hours per aircraft, the following initial spares quantities are recommended for a five-aircraft fleet supporting two years of operations. Actual quantities will be finalized during the Logistics Support Analysis (LSA) phase after First Article approval.

Table 4-2. Recommended Initial Spares for Five-Aircraft Fleet

Part Number	Nomenclature	Qty per A/C	Fleet Spares (5 A/C)	Rationale
QEO-TUR-1000-A	Stabilized Multi-Sensor Turret Assembly	1	1	High cost; MTBF supports low stock
QEO-MWIR-640-B	MWIR InSb FPA Module, 640×512	1	2	Cryocooler-limited life item
QEO-FPA-COOL-A	Closed-Cycle Cryocooler	1	3	Wear-out item; 5 500 h MTBF
QEO-LRF-40K-D	Laser Rangefinder / Designator	1	2	Optical alignment critical
QEO-VPU-10G-C	Video Processing Unit	1	2	Complex electronics
QEO-PCU-850-B	Power Conditioning Unit	1	2	Single-point power path
QEO-SLIP-RING	Fiber-Optic Rotary Joint / Slip Ring	1	3	Rotating interface wear
QEO-FAN-48V-A	Cooling Fan Assembly	2	6	Consumable; high duty cycle
QEO-SPARES-KIT1	30-Day Spares Kit	1	5	Organizational level stock

4.3 Provisioning Documentation. The contractor shall deliver a complete Provisioning Parts List (PPL) in accordance with MIL-STD-1388-2B (or equivalent GEIA-STD-0007) format within 90 days after First Article approval. The PPL shall identify every item down to the piece-part level that is expected to be replaced during organizational or intermediate maintenance.

5. SOFTWARE AND FIRMWARE REQUIREMENTS

5.1 General. All software and firmware resident in the Payload shall be developed in accordance with a tailored application of DO-178C DAL C for flight safety-related functions and DAL D for mission functions. The contractor shall maintain a Software Configuration Index and deliver source code, build scripts, and executable images under the Contract Data Requirements List (CDRL).

5.2 Operating System and Middleware. The Video Processing Unit shall run a real-time operating system (RTOS) that provides deterministic scheduling with worst-case interrupt latency not exceeding 15 μ s. Approved RTOS candidates include VxWorks 7, Integrity-178, or Linux with PREEMPT_RT patch, subject to Government approval of the safety case.

5.3 Cybersecurity. The Payload shall implement the cybersecurity controls identified in CBP-ARCH-2022-03 Appendix D. At a minimum this includes: (a) mutual authentication of all Ethernet command links using certificates issued by the CBP Public Key Infrastructure; (b) encryption of all video and telemetry streams using AES-256; (c) secure boot with measured boot attestation; and (d) a hardware root of trust that prevents unauthorized firmware loading. The contractor shall provide a System Security Plan (SSP) and perform a Security Assessment in accordance with NIST SP 800-53 Rev 5, moderate baseline.

5.4 Software Version Identification. Every executable load shall embed a unique version identifier that is readable via the BIT interface and via a dedicated maintenance connector. The format shall be Major.Minor.Patch-Build (example: 3.2.1-0047). Field updates shall be possible via the Ethernet maintenance port without removing the Payload from the aircraft.

6. ENVIRONMENTAL QUALIFICATION

The Payload shall be designed and qualified to the environmental conditions summarized in Table 6-1. Qualification testing shall follow the procedures of MIL-STD-810H and RTCA DO-160G as indicated. The contractor shall submit a detailed Environmental Qualification Test Plan (EQTP) for Government approval not later than 60 days after contract award.

Table 6-1. Environmental Qualification Requirements

Condition	Requirement	Test Method / Standard
Operating Temperature	–40 °C to +55 °C	MIL-STD-810H Method 501.7 / 502.7
Storage Temperature	–55 °C to +70 °C	MIL-STD-810H Method 501.7 / 502.7
Altitude (operating)	Up to 65 000 ft MSL	MIL-STD-810H Method 500.6
Humidity (operating)	Up to 95 % RH non-condensing	MIL-STD-810H Method 507.6
Vibration (operating)	Aircraft profile per CBP-SPEC-VIB-03	MIL-STD-810H Method 514.8
Shock (functional)	20 g, 11 ms half-sine, 3 axes	MIL-STD-810H Method 516.8
Sand & Dust	Blowing dust, 10.6 g/m ³	MIL-STD-810H Method 510.7
Salt Fog	5 % NaCl, 48 h exposure	MIL-STD-810H Method 509.7
EMI / EMC	MIL-STD-461G, CE102, CS101, RE102, RS103	MIL-STD-461G
Lightning Indirect Effects	DO-160G Section 22, Level 3	RTCA DO-160G
Explosive Atmosphere	Category E (non-sparking)	DO-160G Section 9

6.1 Combined Environments. In addition to the single-condition tests above, the contractor shall perform a combined temperature-altitude-humidity test profile representative of a 30-hour mission at 55 000 ft followed by rapid descent to sea-level tropical conditions. The profile shall be agreed with the Government and documented in the EQTP.

7. VERIFICATION AND VALIDATION

7.1 Verification Methods. Each requirement in this specification shall be verified by one or more of the following methods: Inspection (I), Analysis (A), Demonstration (D), or Test (T). The verification matrix in Table 7-1 allocates the

primary method for selected critical requirements. A complete Verification Cross-Reference Matrix (VCRM) shall be delivered as part of the Systems Engineering Management Plan.

Table 7-1. Extract of Verification Cross-Reference Matrix

Req. ID	Requirement Summary	I	A	D	T
3.1-01	Visible narrow FOV 0.35°		X		X
3.1-04	MWIR NEdT ≤ 25 mK				X
3.1-07	Laser range ≥ 40 km				X
3.1-09	Stabilization ≤ 5 μrad				X
3.1-11	Endurance ≥ 30 h		X		
3.1-13	Power ≤ 850 W steady				X
3.1-15	Mass ≤ 68 kg	X			
3.2-01	MIL-STD-704F power				X
3.3-01	MTBCF ≥ 2 500 h		X		
6.1-01	Operating temp range				X

7.2 First Article Test. A First Article Test (FAT) shall be performed on the first production-representative unit. The FAT shall include full functional performance verification against Table 3-1, environmental qualification per Table 6-1 (or analysis where permitted), and electromagnetic compatibility testing. The Government reserves the right to witness all FAT events.

7.3 Acceptance Test Procedure. Every production unit shall undergo an Acceptance Test Procedure (ATP) that verifies critical performance parameters, BIT functionality, and interface compliance. The ATP shall be approved by the Government prior to first use. A Certificate of Conformance (CoC) shall accompany each delivered unit.

8. QUALITY ASSURANCE PROVISIONS

8.1 Quality Management System. The contractor shall maintain a Quality Management System certified to SAE AS9100D (or equivalent) and shall flow down applicable quality requirements to all subcontractors and suppliers of critical items listed in Table 4-1.

8.2 Configuration Management. All design artifacts, software baselines, and manufacturing process documentation shall be placed under formal configuration control. Changes affecting form, fit, function, or interfaces shall be submitted as Engineering Change Proposals (ECPs) and require Government approval before implementation on deliverable hardware.

8.3 Inspection and Audit Rights. The Government reserves the right to conduct quality audits, process reviews, and source inspections at the contractor's and major subcontractors' facilities with five working days' notice. The contractor shall provide reasonable facilities and access to records.

9. PACKAGING, HANDLING, STORAGE, AND TRANSPORTABILITY

The Payload and all LRUs shall be packaged in reusable transit cases that protect against the storage environments listed in Table 6-1. Cases shall include desiccant, humidity indicator cards, and shock watches. The complete system in its transit case shall not exceed 120 kg and shall be capable of being handled by a two-person lift team. Marking shall comply with MIL-STD-129R. Each case shall be labeled with the part number, serial number, contract number, and a unique barcode readable by standard CBP logistics systems.

10. NOTES

10.1 Intended Use. The Payload is intended for persistent wide-area surveillance in support of border security, maritime domain awareness, and related homeland-security missions. It is not intended for use on manned aircraft without additional certification.

10.2 **Definitions.** For the purposes of this specification the following definitions apply: (a) Critical Failure – any failure that prevents the Payload from performing its primary detection or geo-location function; (b) Organizational Level – maintenance performed by Government or contractor personnel at the operating location using only the supplied tool kit and spares; (c) CEP50 – Circular Error Probable, the radius of a circle centered on the true location that contains 50 % of the measured points; (d) NEdT – Noise Equivalent differential Temperature, a measure of thermal sensitivity of an infrared sensor.

10.3 **Revision History.** Rev A – Initial release (12 Mar 2024). Rev B – Updated power and mass allocations; added SWIR range requirement (28 May 2024). Rev C – Incorporated industry comments; expanded BOM table; added software/cybersecurity section; clarified verification methods (15 Jul 2024).

10.4 **Point of Contact.** Technical questions regarding this specification should be directed to the Contracting Officer's Technical Representative (COTR) identified in the solicitation. Formal requests for clarification or deviation must be submitted through the Contracting Officer.

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